



Stand-Cornter: An End-to-End Automatic Stand Count for Seedling Maize Using YOLOv9

Dewi Endah Kharismawati and Toni Kazic

Department of Electrical and Computer Engineering, University of Missouri



Accurate stand counts are essential for crop management and research, yet the traditional method of counting plants by walking through fields is labor-intensive, slow, and prone to errors. To address this, we propose *Stand-Cornter*, a robust end-to-end algorithm designed to provide accurate stand counts for maize fields from imagery captured by freely flown, inexpensive drones and affordable computer processing. *Stand-Cornter* operates in two modes: counting from raw frames with homography matrices and from fragmented mosaics, both fed to the YOLOv9 model. The model is trained to recognize seedling maize plants in the V2–V10 growth stages, either singly or in groups of two and three plants. We evaluated *Stand-Cornter* against in-field stand counts across various maize lines in our 2024 summer nursery.

Dataset

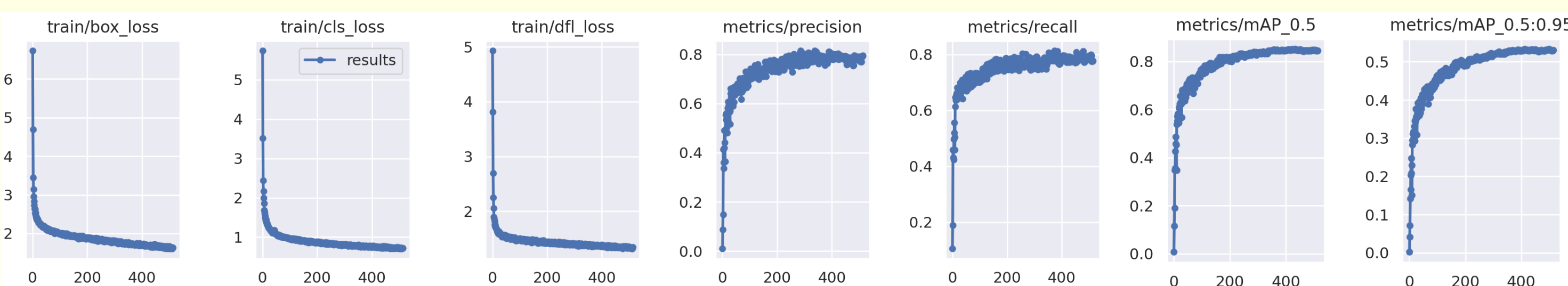
- YOLOv9 Training data from 2019, 2020 and 2021 field seasons
- YOLOv9 Test data from 2022 field season
- Stand-Conter* test data from 2024 field season



Sample of training data



Sample of testing data

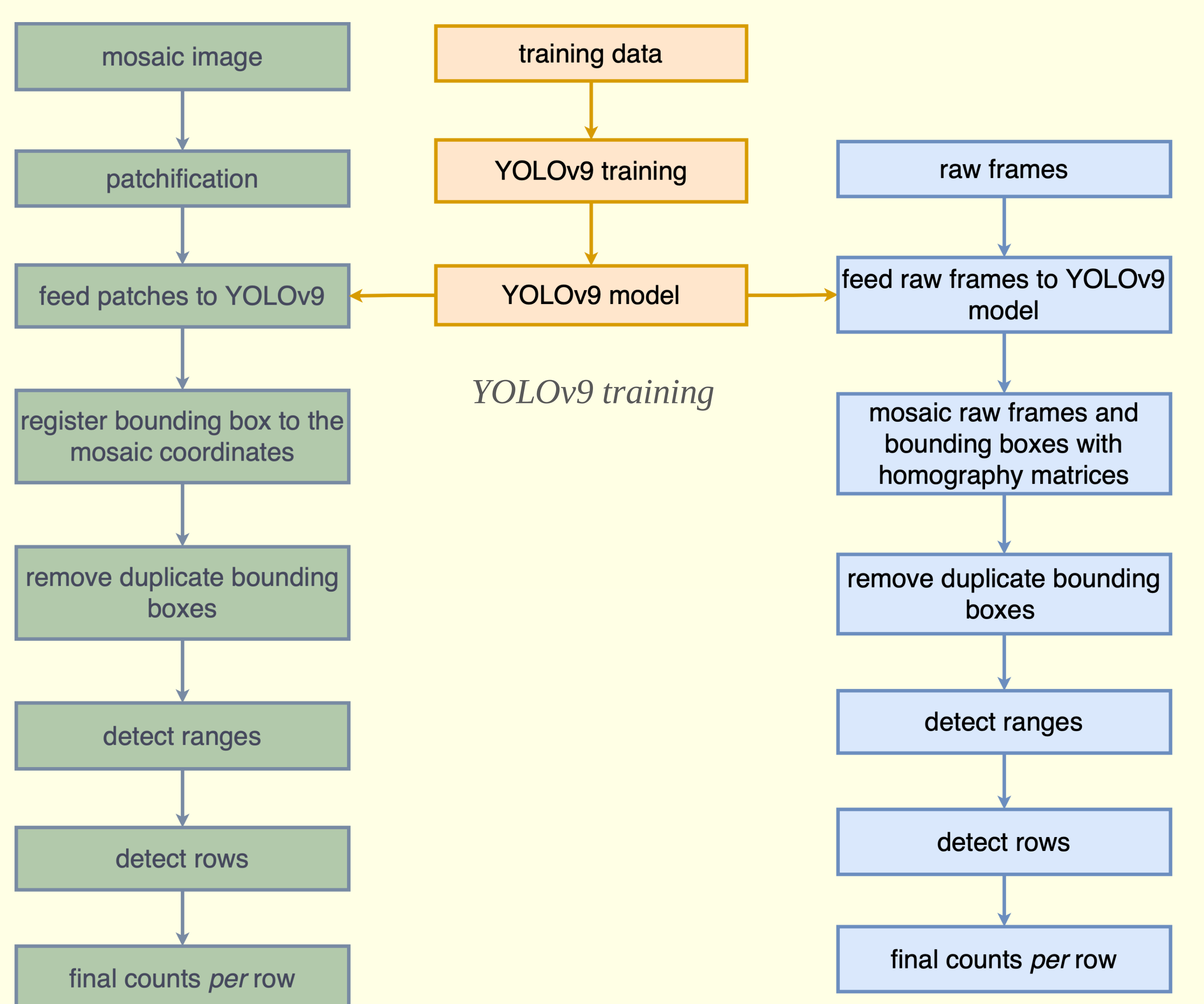


YOLOv9 training metrics

Classes	Images	No. Plants	Precision	Recall	mAP
All Plants	175	10438	0.72	0.75	0.78
One Plant	175	9342	0.91	0.82	0.92
Two Plant	175	861	0.56	0.73	0.65
Three Plants	175	235	0.69	0.70	0.76

YOLOv9 evaluation for different classes of plant groupings in YOLO test data

Pipeline



Stand-Cornter with Mosaics

Stand-Cornter with Raw Frames

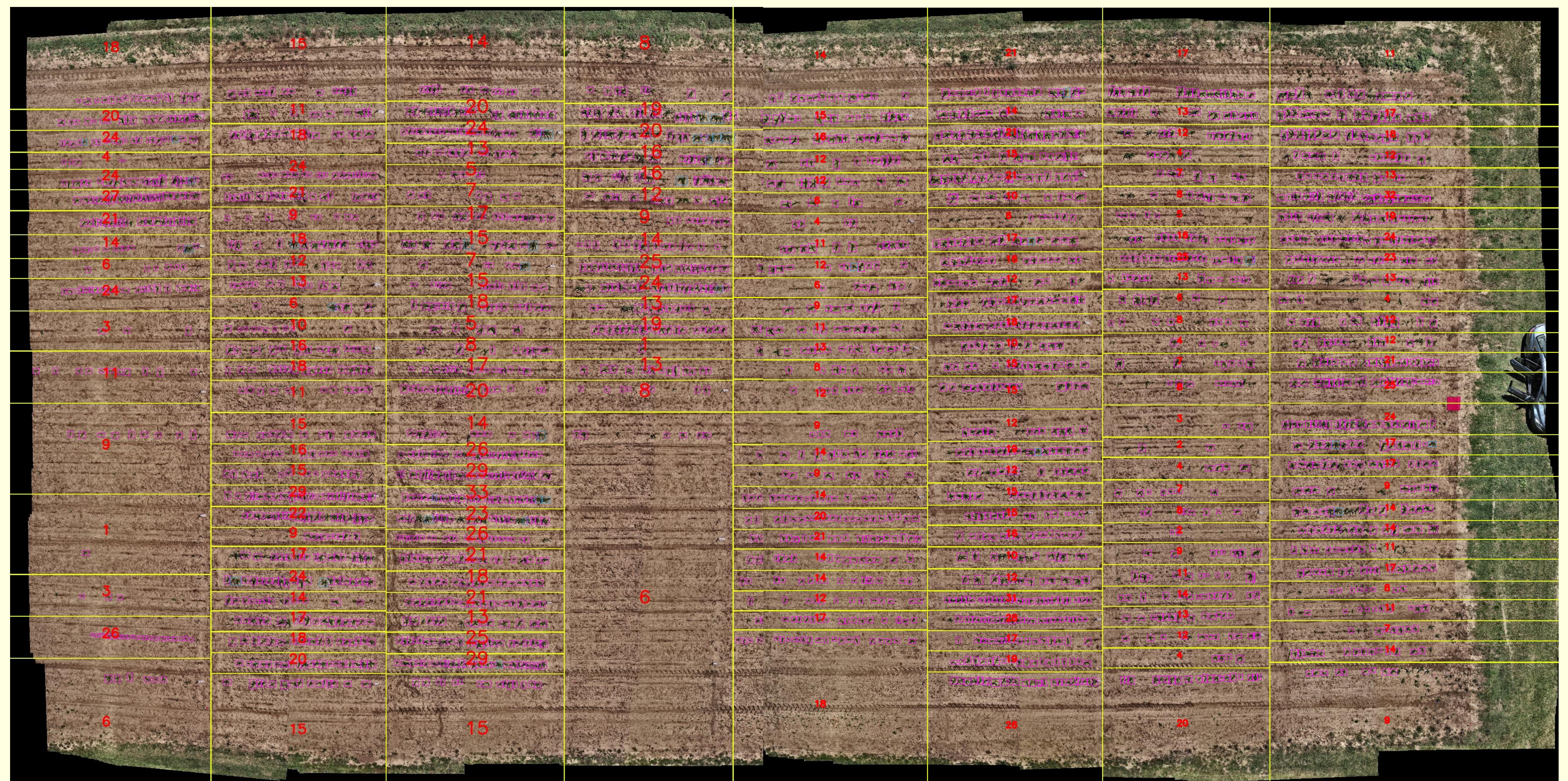
Warping bounding boxes with homography matrices

$$H_{1 \leftarrow i} = H_{1 \leftarrow 2} \times H_{2 \leftarrow 3} \times \dots \times H_{i-1 \leftarrow i}$$

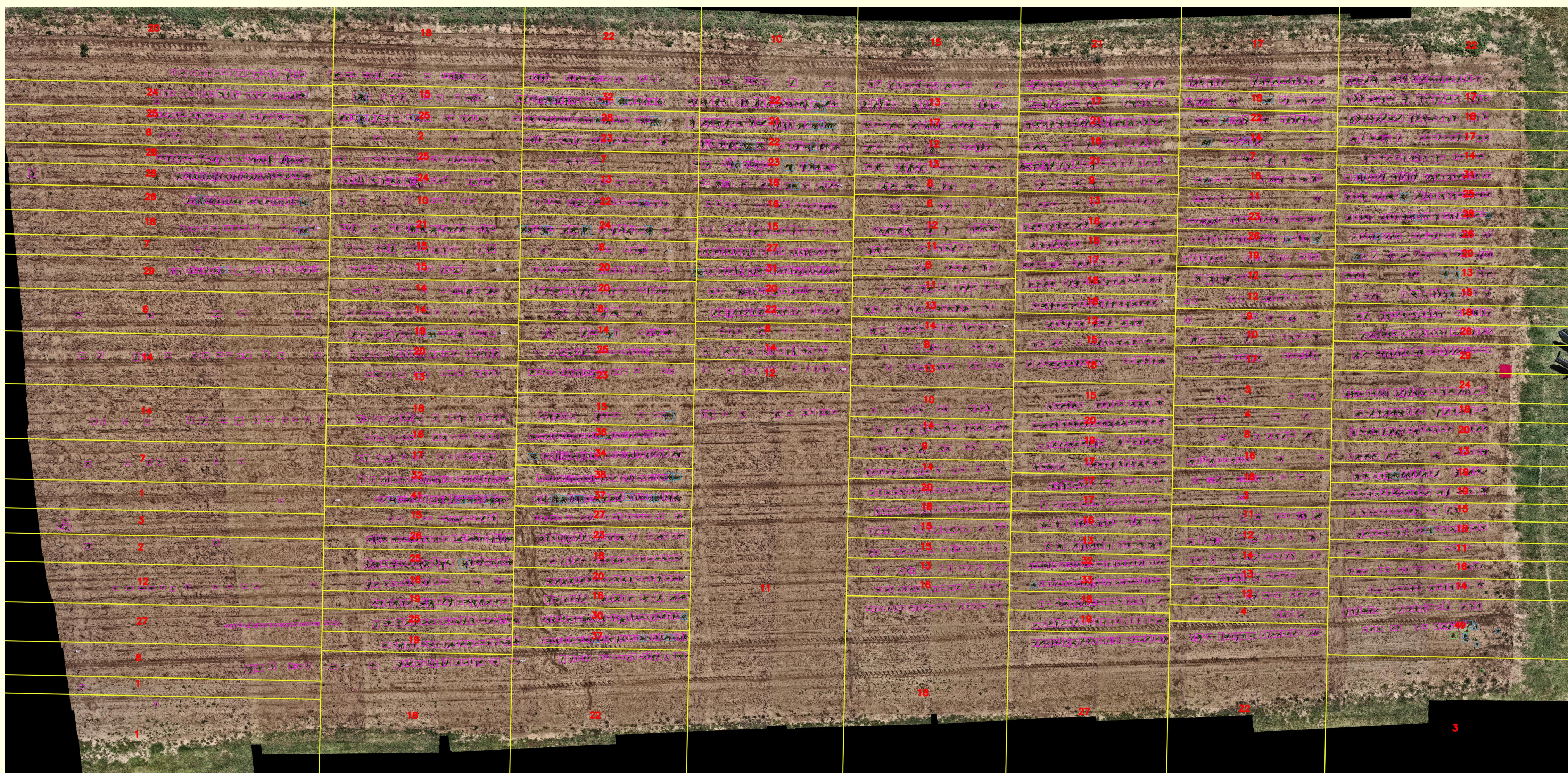
where $H_{1 \leftarrow i}$ is the cumulative homography from frame 1 to frame i and w is the warp factor:

$$\begin{bmatrix} x'_i \\ y'_i \\ w \end{bmatrix} = H_{1 \leftarrow i} \begin{bmatrix} x_i \\ y_i \\ 1 \end{bmatrix}$$

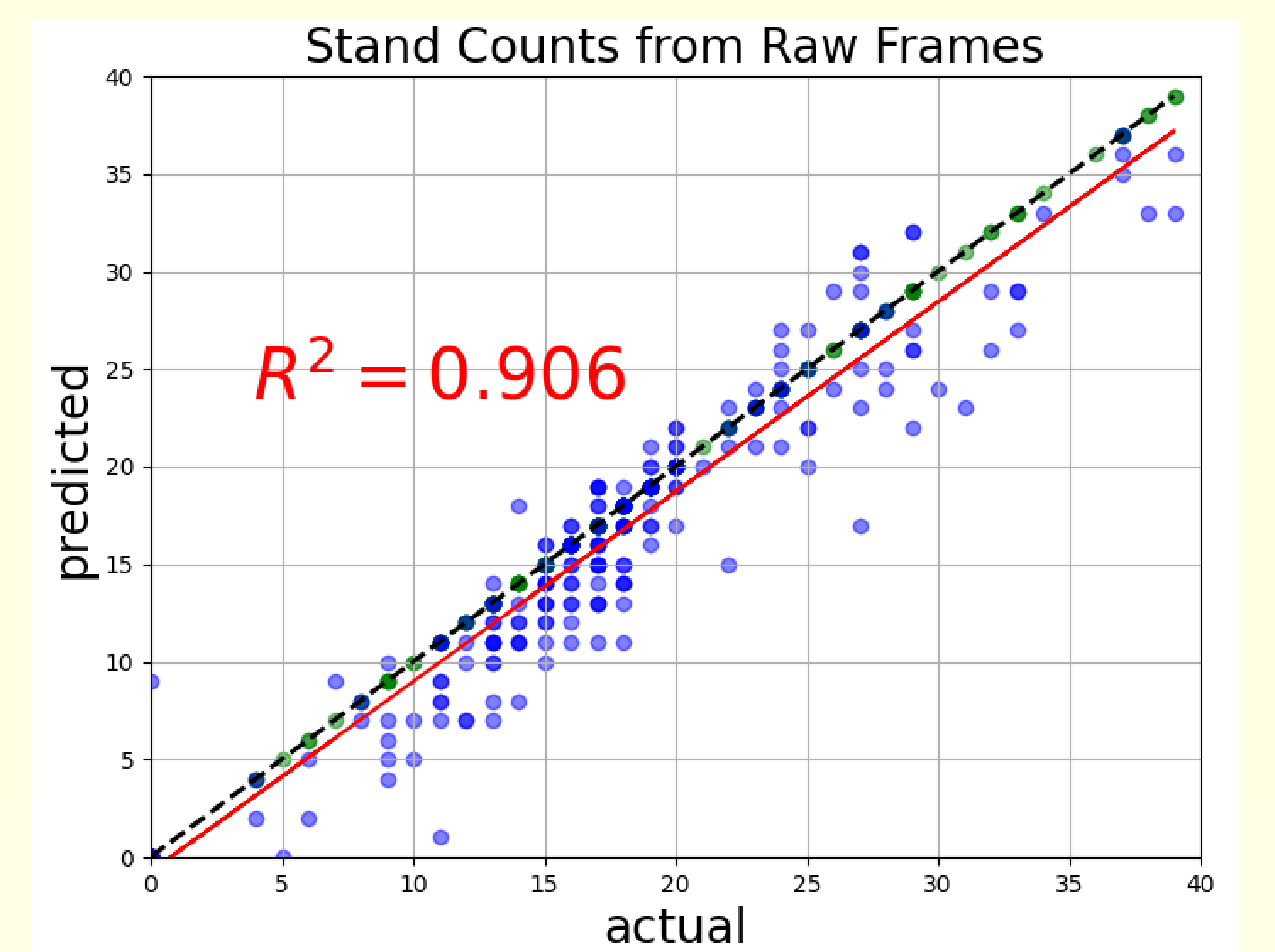
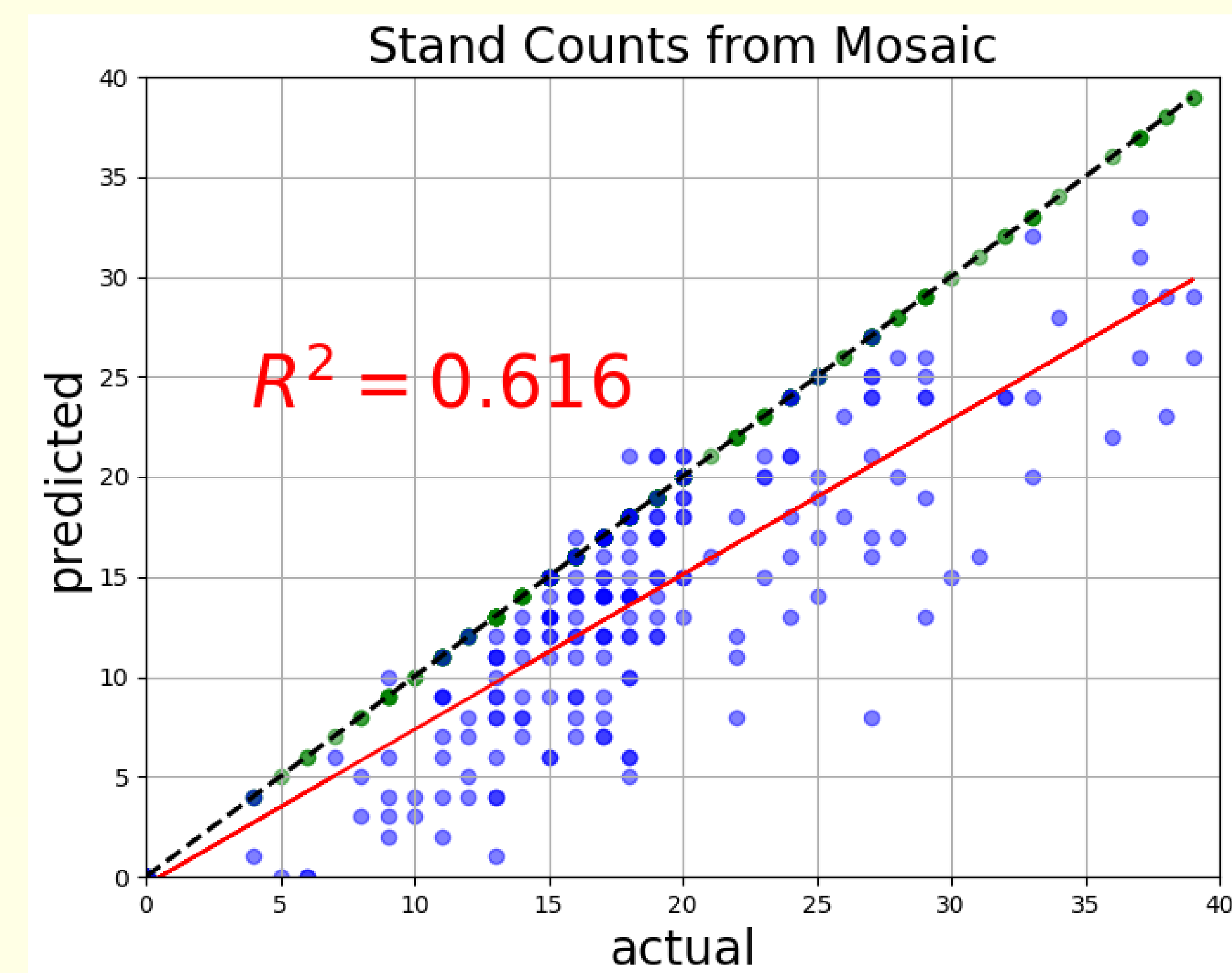
Results



Stand counts from mosaic image



Stand counts from raw frames



Discussion

- It achieves an R^2 of 0.906 for raw frames and 0.616 for mosaics.
- Counting from raw frames yields higher accuracy because a single plant can appear in multiple frames with 90% overlap. If detection fails in one frame, there's a possibility it will be detected in another. Duplication is expected and is filtered out using the strongest confidence value.
- Stand-Cornter* reliably distinguishes maize from weeds and other grasses.
- It also detects rows and ranges based on detected bounding boxes, enabling row-based counting.
- The lightweight network processes 83 full-sized 4K frames in 60.63 seconds, with 49.42 seconds for running YOLOv9 and 11.21 seconds for assembling the global bounding boxes, detecting ranges and rows, and counting.
- It is suitable for real-time deployment on cost-effective platforms.

References

- Wang, C. Y., Yeh, I. H., & Liao, H. Y. M. (2024). YOLOv9: Learning what you want to learn using programmable gradient information. *arXiv preprint arXiv:2402.13616*.
- Kharismawati, D. E., Kazic, T. (2024). *Maizaic*: a robust end-to-end pipeline to mosaic freely flown aerial RGB video. Manuscript in preparation